Assignment 8: Agent-Based Models

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Exercise 1

```
results_ex1 <- run_abm()
```

i. What are the columns called that hold the counts of susceptible, infectious, and recovered people?

I, S, and R.

ii. How many agents (i.e. people) were in this simulation in total?

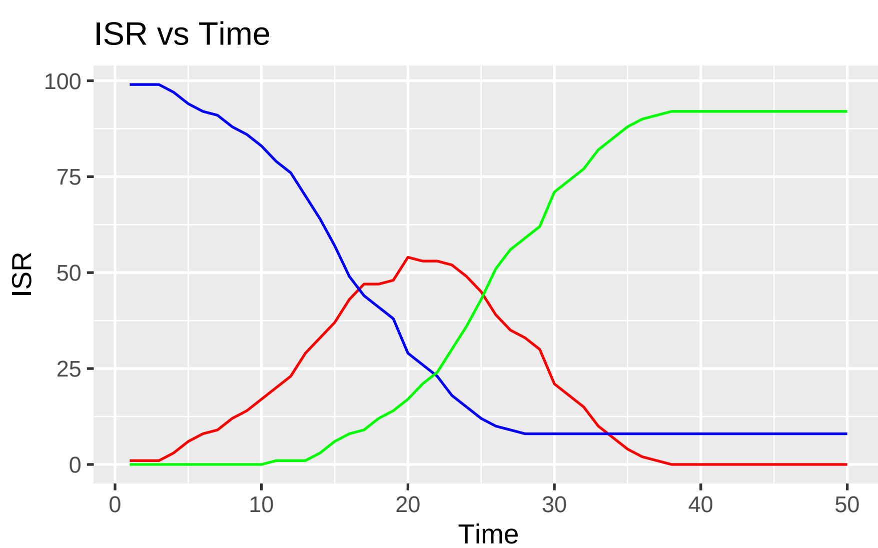
100 people.

iii. How many time steps did the simulation run for?

50 steps.

Exercise 2

```
ggplot(results_ex1)+
  geom_line(aes(x=time, y=I), color='red')+
  geom_line(aes(x=time, y=S), color='blue')+
  geom_line(aes(x=time, y=R), color='green')+
  labs(title="ISR vs Time", x="Time", y="ISR")
```

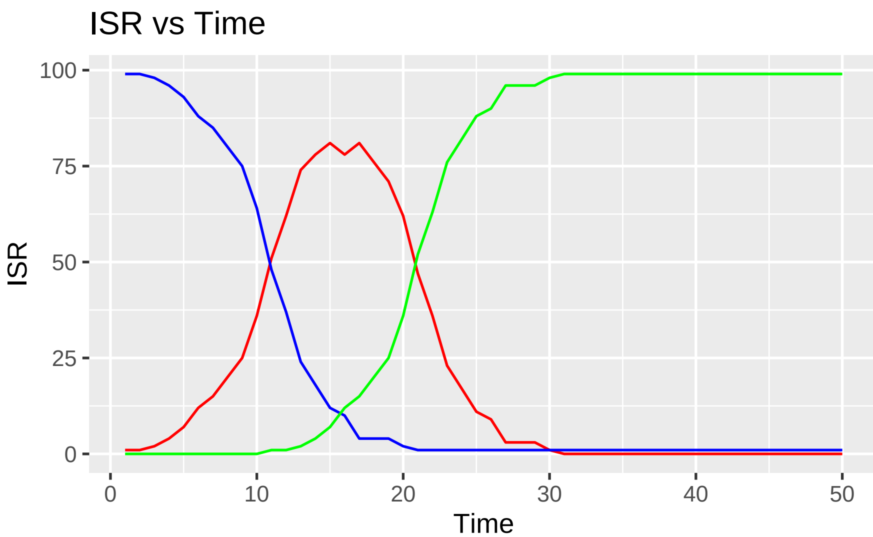


Exercise 3

```
plot_sir <- function(.data, title){
  ggplot(.data) +
    geom_line(aes(x=time, y=I), color='red') +
    geom_line(aes(x=time, y=S), color='blue') +
    geom_line(aes(x=time, y=R), color='green') +
    labs(title = "ISR vs Time", x = "Time", y = "ISR")
}
```

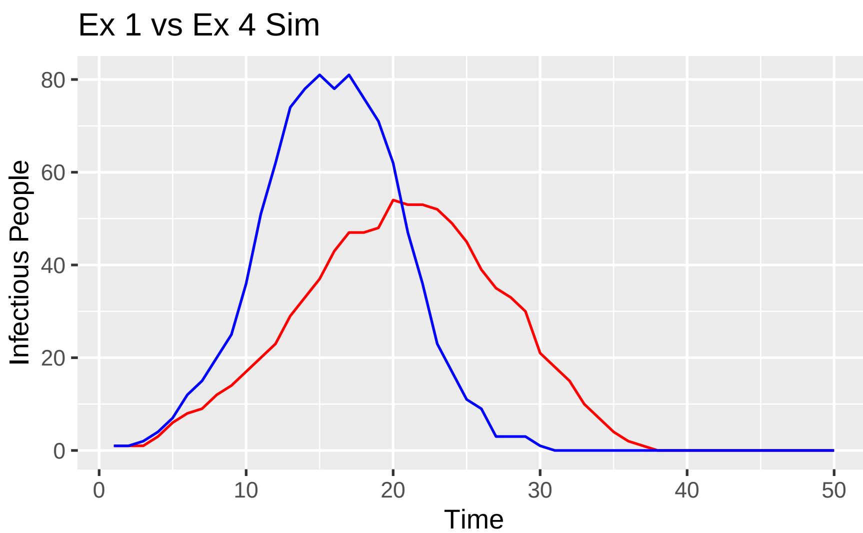
Exercise 4

```
results_ex4 <- run_abm(base_prob_infection = 0.1)
plot_sir(results_ex4, "ISR vs Time with Infection Probability = 0.1")
```



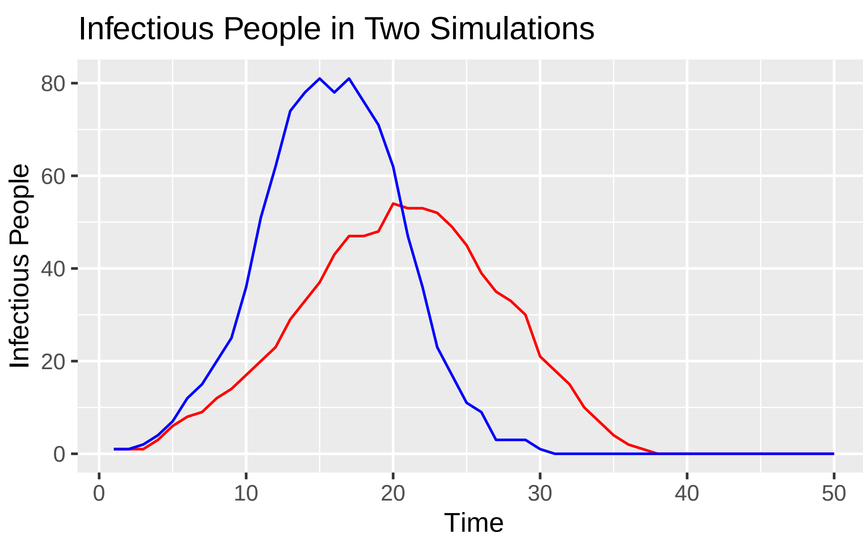
Exercise 5

```
ggplot() +
  geom_line(data = results_ex1, aes(x = time, y = I), color = 'red') +
  geom_line(data = results_ex4, aes(x = time, y = I), color = 'blue') +
  labs(title = "Ex 1 vs Ex 4 Sim", x = "Time", y = "Infectious People")
```



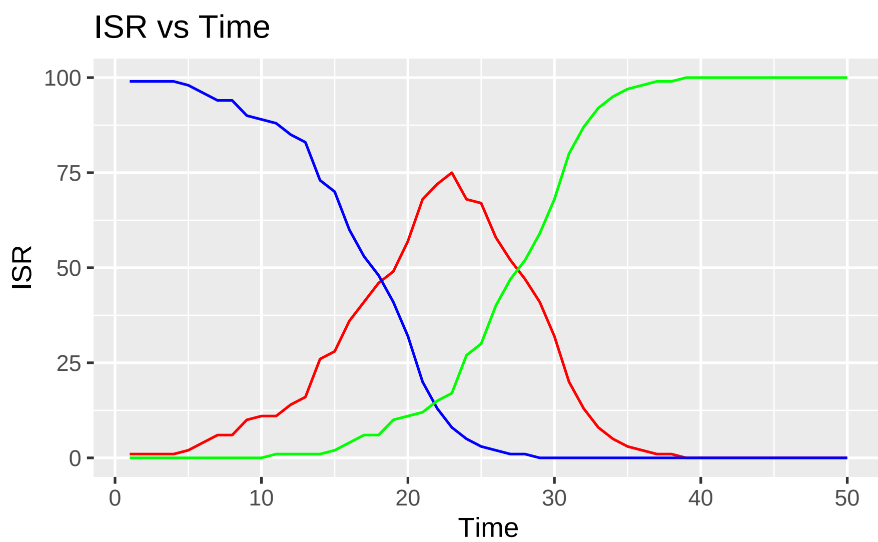
(Simulation 1 is RED and Simulation 2 is BLUE) Simulation 1 has a lower peak for the number of people infected, and simulation 2 has a higher peak. So this means that sim 1 had less people infected at once than sim 2. However, it took longer for the number of infected people to decrease in sim 1. In sim 1, it took about 38 days for the number of infected people to get to zero. But it took sim 2 only 30 days.

```
plot_comparison <- function(df1, df2, title){
  ggplot() +
  geom_line(data = df1, aes(x = time, y = I), color = 'red') +
  geom_line(data = df2, aes(x = time, y = I), color = 'blue') +
  labs(title = title, x = "Time", y = "Infectious People")
}
plot_comparison(results_ex1, results_ex4, "Infectious People in Two Simulations")
```

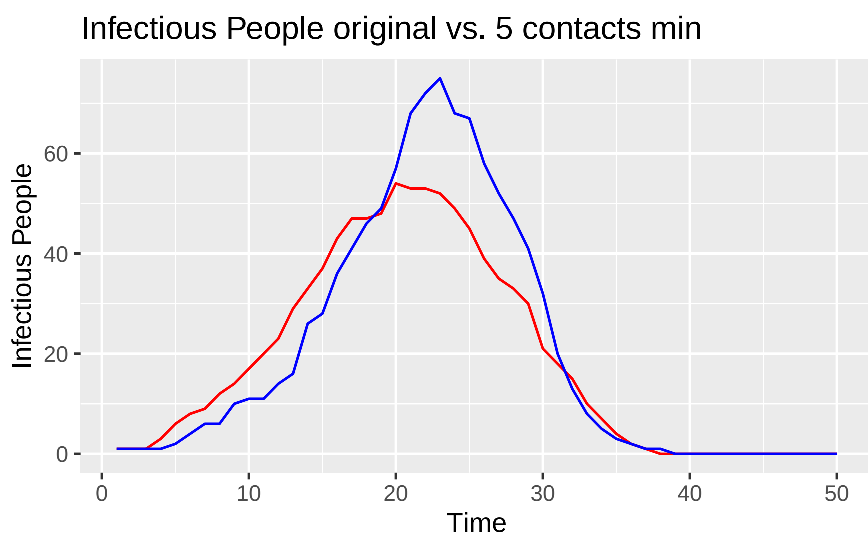


Exercise 6

```
results_ex6 <- run_abm(min_interactions = 5)
plot_sir(results_ex6, "SIR Model Minimum of 5 Contacts")
```



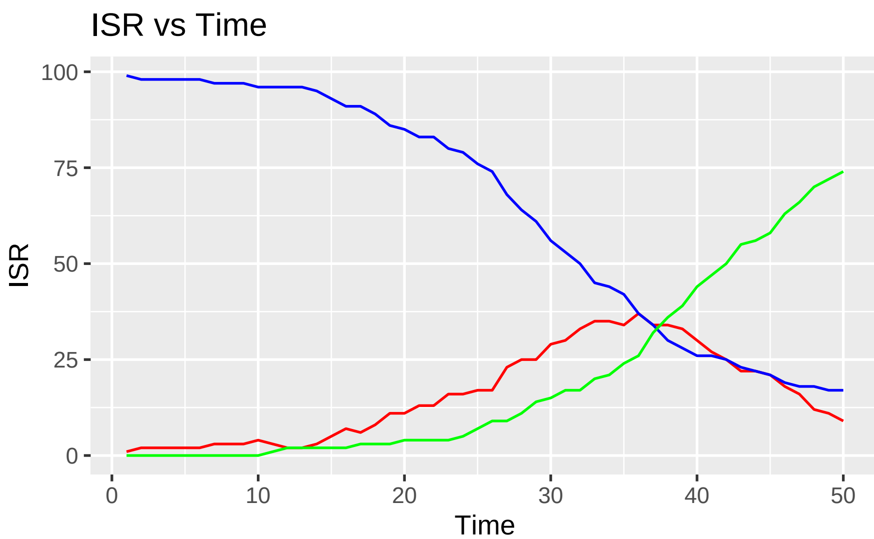

```
plot_comparison(results_ex1, results_ex6, "Infectious People original vs. 5 contacts min")
```



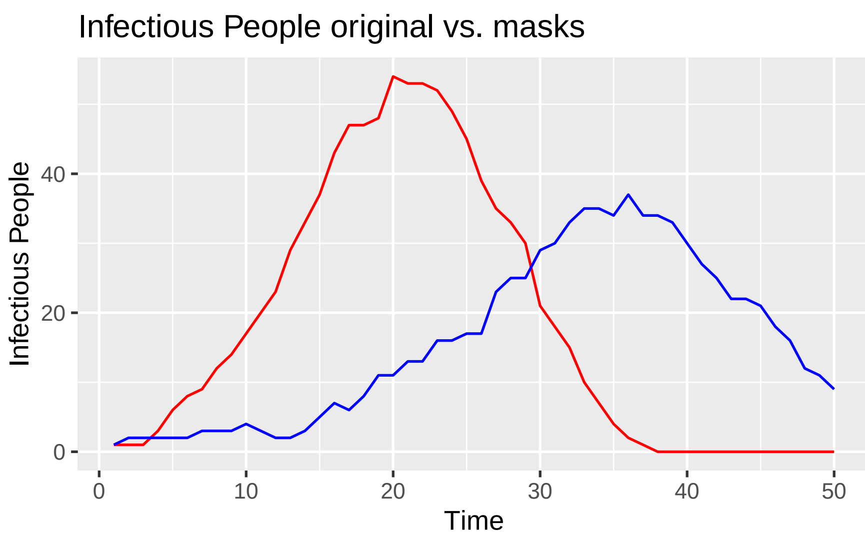
This sim 3 is different then the last 2. First, its infected graphs peak is higher than sim 1 and lower than sim 2. Secondly, its suspected graph reaches zero before sim 1 and after sim 2 when their suspected num of people became 0. And finally, the number of recovered people graph is the same as the suspected but just reversed. In sim 1, the peak is reached at around 23 days. In sim 2 the peak is reached at around 16 days. And in sim 3 the peak is reached at 23 days.

Exercise 7

```
results_ex7 <- run_abm(masking = TRUE, masking_prob = 0.5, mask_effectiveness = 0.3)
plot_sir(results_ex7, "SIR Model with Masks")
```



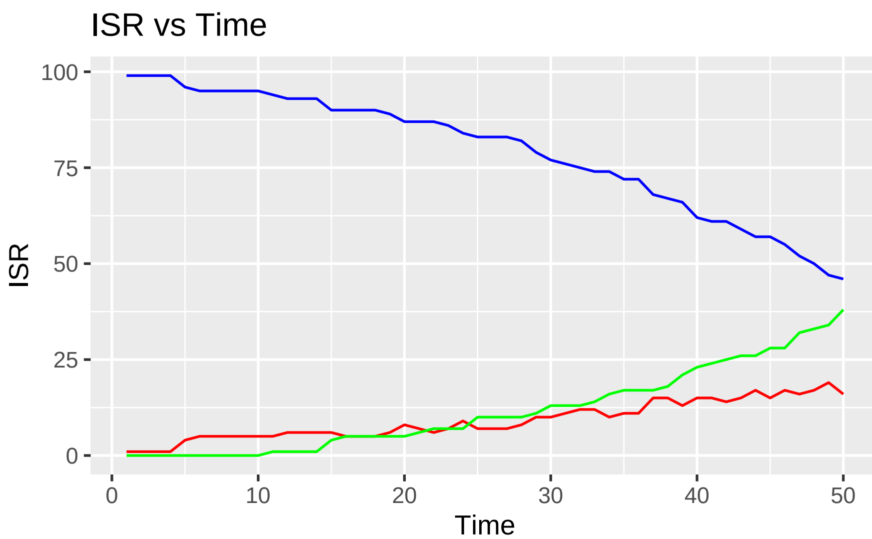

```r
plot_comparison(results_ex1, results_ex7, "Infectious People original vs. masks")
```



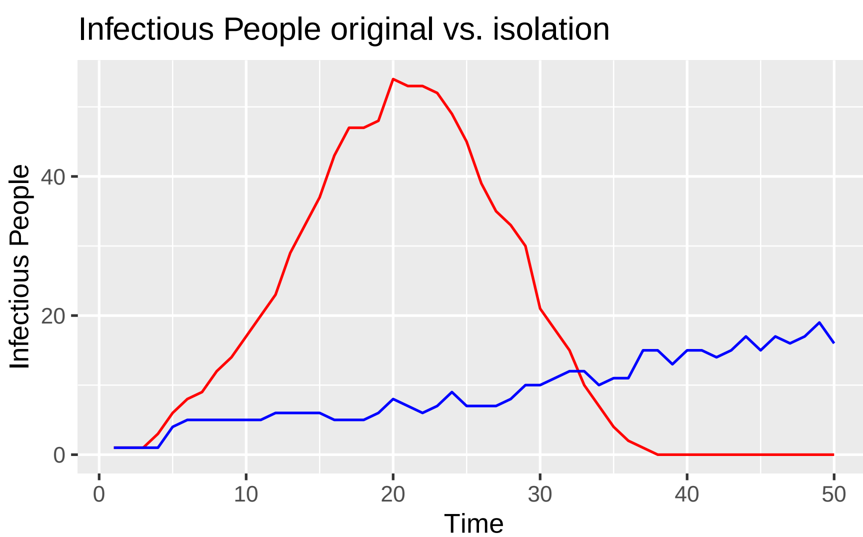
With masks the disease takes way longer to go around. The peak for sim 4 happens way later than all the other sims.

Exercise 8

```r
results_ex8 <- run_abm(isolation = TRUE)
plot_sir(results_ex8, "SIR Model with Isolation")
```




```
plot_comparison(results_ex1, results_ex8, "Infectious People original vs. isolation")
```



With isolation the disease takes way longer to go around. The peak for sim 5 happens way later than all the other sims.

Exercise 9

The results from the different simulations show how various factors impact disease spread. In Simulation 1, the peak number of infections was lower than in Simulation 2, meaning fewer people got infected at the same time. However, it took longer for the infection rate to drop in Simulation 1, about 38 days to hit zero compared to 30 days for Simulation 2. Simulation 3 had a peak that was higher than Simulation 1 but lower than Simulation 2. It reached zero susceptible individuals before Simulation 1 but after Simulation 2. When masks were introduced in Simulation 4, the spread slowed down, with the peak happening later. Similarly, Simulation 5, which involved self-isolation, showed an even longer spread time. These variations highlight the importance of public health measures like masking and isolation to reduce infection peaks and ease the burden on healthcare systems.

To improve the model, we could add more factors that affect disease spread, like vaccination, population density, or movement patterns. It would also be great to compare our model's predictions with real outbreak data to see how accurately it reflects actual infection rates. This way, we can make the model a more effective tool for understanding and controlling infectious diseases.